

Brhane Teamrat Gidey

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RESEARCH INTERESTS

AI for Software Engineering · Deep learning · Computer vision · Natural language processing · Generative AI · Agentic AI systems

PROFESSIONAL SUMMARY

Software and AI Engineer with over five years of industry experience, currently completing an MSc at the University of Calabria with a research thesis at York University, Canada. Top Rated on Upwork with a 100% job success score and five-star rated on Freelancer.com, with AI and full-stack systems delivered for clients across France, the USA, Canada, Italy, the Philippines and Saudi Arabia. I specialise in deep learning, generative AI, PyTorch, LLMs and agentic AI. I am seeking a PhD position to pursue rigorous research at the intersection of artificial intelligence and software quality assurance.

EDUCATION

M.Sc. in Artificial Intelligence and Computer Science · *University of Calabria, Italy* Oct 2024 – Aug 2026 (Expected)

Research thesis at York University, Canada · Supervisors: Prof. Song Wang (York) and Prof. Francesco Calimeri (UniCal)

- Thesis: Contract-Guided LLM Code Generation.
- Coursework: Intelligent Agents, Deep Learning, Optimization for Machine learning, Intelligent Systems, Statistical Methods for Data Science, Neurosymbolic AI, Massively Parallel Programming on GPUs, Big Data Analytics and Reasoning.

B.Sc. in Computer Science and Engineering · *Mekelle University, Mekelle Institute of Technology, Ethiopia* Oct 2014 – Jul 2019

- Thesis: Emotion-Based Music Player, a real-time facial expression recognition system that selects music based on the user's detected emotion using computer vision and ML classification.
- Coursework: Algorithms, Data Structures, Artificial Intelligence and fuzzy logic, Neural Networks, Distributed Systems, Software Engineering, Database Management.

RESEARCH IN PROGRESS

Contract-Guided LLM Code Generation · *MSc Thesis, York University and University of Calabria*

Supervisors: Prof. Song Wang (York University) and Prof. Francesco Calimeri (University of Calabria)

This work asks whether small LLMs augmented with formal contracts can match the functional correctness of much larger frontier models. A Python DbC runtime is being built to enforce preconditions, postconditions and class invariants on generated code. Contracts are synthesised by an LLM, validated for completeness and consistency, and then used to guide a five-stage pipeline culminating in iterative contract-driven refinement and re-verification. Evaluation runs on HumanEval, LiveCodeBench and BigCodeBench, comparing contract-guided small models against frontier baselines.

- Implementing the Python DbC runtime using a decorator-based approach for pre and postcondition enforcement and invariant checking, mirroring the contract tooling of OpenJML and native Eiffel.
- Testing smaller models including DeepSeek-Coder-33B, Qwen2.5-32B, GPT-3.5, Claude 3 Sonnet and Gemini 1.5 Pro against frontier baselines such as DeepSeek-V3, GPT-4, Claude 4.6 and Gemini 3.

PROJECTS

Medical Knowledge and Healthcare AI Assistant | *RAG · LLM · Agentic AI · FAISS*

- Built a retrieval-augmented generation system grounded in WHO and CDC documents, combining FAISS vector search with a multi-step LLM reasoning layer to answer clinical queries with cited sources.

Automatic Medical Imaging Report Generator | *Transformer · ResNet-50 · GPT-2 · Computer Vision*

- Developed a model that generates clinical radiology reports from chest X-ray images by fusing ResNet-50 visual features with GPT-2 text generation, covering the full pipeline from data preparation through benchmark evaluation.

LLM-Based Job Recommendation System | *LLM · Semantic Embeddings · NLP*

- A recommendation engine that uses LLM semantic embeddings to match candidates to roles based on meaning rather than keyword overlap, capturing nuanced skill-to-role relationships.

Personal Chef AI | *Multi-Agent LLM · Agentic Orchestration · Prompt Engineering*

- A multi-agent system that generates personalised recipes and step-by-step cooking guidance adapted to the user's dietary preferences and available ingredients.

Integrated EMR and Laboratory Information System | [Spring Boot](#) · [React](#) · [Healthcare](#)

- Full-stack system integrating patient records with laboratory workflows, delivered for a production healthcare client and currently in active clinical use.

Electronic Prescription and Pharmacy Management System | [Spring Boot](#) · [React](#) · [Flutter](#) · [Laravel](#)

- Secure digital prescription platform connecting healthcare providers and pharmacies, with a companion backend for inventory and order management.

Food Order and Delivery Application | [React Native](#) · [Flutter](#) · [Spring Boot](#) · [Real-Time Tracking](#)

- End-to-end food ordering app with real-time delivery tracking, covering both the frontend experience and the backend order processing infrastructure.

PROFESSIONAL EXPERIENCE

Software and AI Engineer · [Upwork \(Top Rated, 100% Job Success\)](#) · [Freelancer.com \(5-Star\)](#) Dec 2022 – Present

- Designed and deployed RAG pipelines, LLM-integrated assistants and computer vision systems for clients in healthcare, productivity and logistics across more than seven countries.
- Built full-stack enterprise systems with microservices and multi-tenant architectures, maintaining a 100% job success score across more than twenty engagements.

Software Engineer · [Medco Technology Solutions](#) · [Addis Ababa, Ethiopia](#) Mar 2020 – Sep 2023

- Led development of healthcare and government software from requirements through production, mentored junior engineers and delivered an integrated EMR and laboratory system now in active clinical use.

ICT Consultant · [Nayna Saving and Credit Cooperative](#) · [Ethiopia](#) Sep 2019 – Mar 2020

- Advised on IT strategy, oversaw infrastructure, supervised two staff members and led training to improve adoption of internal systems.

TECHNICAL SKILLS

Languages: Python, Java, JavaScript, TypeScript, Dart, PHP

AI and ML: TensorFlow, PyTorch, Hugging Face, LangChain, OpenAI API, FAISS, Deep Learning, Computer Vision, NLP, LLMs, Generative AI, RAG, Agentic AI, AI for Software Engineering

Software Engineering: Design by Contract, Program Verification, Automated Testing, Runtime Monitoring

Backend and Cloud: Spring Boot, FastAPI, REST APIs, Microservices, AWS, Docker

Frontend and Mobile: React, React Native, Flutter, TypeScript

CERTIFICATIONS

- Deep Learning Specialization, deeplearning.ai and Coursera — Andrew Ng
- Natural Language Processing Specialization, deeplearning.ai and Coursera — Andrew Ng
- Machine Learning Engineering for Production (MLOps) Specialization, deeplearning.ai and Coursera — Andrew Ng